

Governing Our Digital Future: Every Government Needs an Open Source Program Office

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The Quiet Revolution

There is a quiet revolution underway in how governments build technology. That revolution does not make headlines and does not involve a single dramatic policy announcement. It is happening incrementally, agency by agency, country by country, and it is reshaping the foundation of digital public services worldwide.

That revolution is the shift to **open source software as the default building block** of Digital Public Infrastructure (DPI).

Governments that will lead digitally in the next decade are not the ones that spend the most on technology. They are the ones who govern it most wisely.

This post examines the urgency of setting up open source governance, the risks of inaction, what an OSPO is, how a government OSPO differs from a corporate one, what it costs to run open source responsibly, how to build an OSPO in phases, and what policy foundations need to be in place to make it successful.

Go deeper with other related posts:

Figure 1: Figure 0: Core DPI Functions

- Beyond Source Code: The Case for Government Open Source Program Offices in the Context of Digital Public Infrastructure
 - Leveraging Open Source to Build Digital Public Infrastructure that is Inclusive, Transparent, and Scalable
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The Foundation Is Already Open Source

DPI is the layer of systems that makes modern government function (Figure 0).

- It is the digital identity platform that verifies a citizen's right to access a service.
- It is the payment system that delivers a social benefit.
- It is the data exchange protocol that allows one ministry to securely share information with another.

These systems are built in layers (Figure 1). At the top sit citizen-facing services: the tax portal, the e-health application, and the digital permits. Beneath them, in the middle layer, sits DPI itself. And beneath DPI, in the bottom layer, sits the software that makes all of it run: operating system, databases, cryptographic libraries, networking tools, security frameworks. That bottom layer is overwhelmingly open source software.

- The Linux kernel is the foundation of virtually every government server, cloud environment, and embedded system worldwide.
- PostgreSQL is the most widely deployed open source relational database in government, used for everything from citizen registries to financial systems and case management platforms.
- OpenSSL is the cryptographic library that secures the majority of government web communications, VPNs, and digital certificate infrastructure.
- Kubernetes is the container orchestration platform that increasingly underpins government cloud infrastructure and digital service delivery at scale. It is adopted by agencies across the US, EU, and beyond.
- The Apache HTTP Server / Nginx is the web server that powers the majority of government-facing portals, APIs, and digital services globally.
- Drupal is the e-content management system of choice for government websites worldwide.
- And the list goes on to cover thousands of open source libraries, frameworks, platforms, and individual components and packages.

Figure 2: Figure 1: Modern DPI is built in layers with open source software as its core (bottom layer)

Side note: Observers will notice that every critical project listed above is hosted and governed by a neutral open source foundation, not by any single company or government. Why that choice matters for digital sovereignty is a story worth telling on its own. I am planning a dedicated post on why that matters.

This heavy reliance on open source is not a preference or a policy choice. It is the technical reality of how modern software is built. The decision to use open source in governments has already been made by developers, procurement officers, vendors, and the architecture of the modern internet itself.

Risks of Unmanaged Open Source

Most governments today are passive consumers of open source.

- Developers download code freely and use and adapt it as they need.
- Agencies adopt frameworks and platforms without central visibility or coordination across them.
- Vendors deliver products built on and with open source components that they may have never disclosed.

The result is the Iceberg Model of IT risk. The software that the government formally procures, the commercial products it licenses, tracks, and manages, is the visible tip. Beneath the surface lies a vast and largely invisible mass: open source components that constitute 80 to 90 percent of the code actually running in government systems.

What are some of the most critical risks?

- **Shadow OSS:** No one knows what is running, where it came from, or whether it has been updated.
- **License violations:** Every piece of open source software carries a license. Using code in violation of its terms creates legal exposure and, in some cases, intellectual property risk.
- **Unpatched vulnerabilities:** Without systematic tracking, critical security patches go unapplied and known vulnerabilities remain open.
- **Security debt:** Each unmanaged component is a liability that accumulates over time. The older and more complex the system, the harder and

Figure 3: Figure 2: Government OSPO vs Corporate OSPO

more expensive it becomes to recover from that technical security debt.

- **Supply chain risk:** The Log4j vulnerability in December 2021 (briefing on this incident courtesy of IBM) was a defining moment. A single widely-used logging library had a critical flaw. Governments (and enterprises) worldwide scrambled for weeks and months trying to answer the questions: *do we use this library, and if so, where and on what version are we?* Organizations that had mapped their software inventory answered those questions in hours. Those who had not faced a very different reality. Their remediation effort ran into months, consumed millions in unplanned costs, and exposed a fundamental truth: *you cannot protect what you cannot see.*

The risks of *unmanaged* open source are active and growing; however, they are entirely preventable.

The Solution: The Government OSPO

The answer to address these risks is an Open Source Program Office (OSPO). Before explaining what an OSPO does, I need to be precise about something that is frequently misunderstood: a government OSPO is fundamentally different from a corporate one (Figure 2).

A corporate OSPO, at a company like Google, Microsoft, or Meta, exists to manage open source risk and use open source as a tool for competitive advantage. Its goals are productivity, efficiency, speed, and innovation that outpace rivals.

A government OSPO has a public mission and a *different* set of goals that include:

- **Digital sovereignty:** ensuring the nation controls its own digital destiny, that no single vendor can hold government services hostage, and that government can audit, modify, and operate its own infrastructure.
- **Public transparency:** ensuring that software built with public funds is accountable to citizens. Open code can be scrutinized, verified, and trusted.
- **Inter-agency collaboration:** breaking down silos that cause ministries to rebuild the same systems repeatedly, at enormous cost to taxpayers.
- **Long-term sustainability:** ensuring that the critical open source projects the government depends on remain healthy, not just for the government's benefit, but for everyone who relies on them.
- **Vendor independence:** the ability to choose the best solution, switch suppliers, and avoid lock-in that compromises both cost and security.

Figure 4: Figure 3: Examples from the global landscape

The shift a government OSPO embodies is from passive consumer to active steward of digital public goods. Germany acted on it when the federal government established ZenDiS (the Center for Digital Sovereignty) with a mandate to coordinate open source strategy across the entire federal administration. Today, the majority of German federal agencies are using open source and operating dedicated OSPOs to manage it. Italy went further by setting bold quantitative targets: 150 public bodies actively publishing open source code, and 3,000 agencies reusing it, all by 2026. They built Developers Italia, a national platform for sharing government code, and enshrined open source as the legal default under their Digital Administration Code (Figure 3).

These are policy commitments backed by law, budget, and institutional authority.

Government OSPO's Core Functions: Control and Stewardship

The functions of a government OSPO fall into two categories: inbound and outbound.

Inbound functions: Governing What the Government Consumes

Policy and compliance are the foundation for the inbound functions. The OSPO establishes the rules:

- A use-first mandate requiring agencies to evaluate existing open source solutions before commissioning new development or procuring proprietary software
- A publish-by-default principle for publicly funded code
- A license compliance framework
- A mandate for Software Bills of Materials, or SBOMs, across all new software acquisition and development

What's an SBOM?

An SBOM is a formal, machine-readable inventory of every open source component in a piece of software. When Log4j was disclosed, organizations with SBOMs knew their exposure within hours.

Related article:

- SBOM Adoption: In Support of Better License Compliance and Software Security Practices

Security and vulnerability management is the operational complement to policy. The OSPO runs the monitoring infrastructure, coordinates patching across agencies, and ensures that a Log4j-scale event triggers a unified government response rather than a fragmented agency-by-agency scramble.

Procurement reform is perhaps the highest-leverage inbound function. Procurement rules and regulations in governments were written for a world of proprietary licenses. They unconsciously disadvantage open source, which follows a completely different economic model. The OSPO fixes this by training procurement officers, developing model Request For Proposal (RFP) templates that evaluate project community health and require vendors to contribute modifications back upstream, and formally recognizing open source software as Commercial Off-The-Shelf (COTS) software that officers are already mandated to consider.

Outbound Functions: Create Public Value

Community engagement means being a responsible steward of the digital commons. Every government depends on critical upstream projects. The OSPO identifies the ten to twenty most essential ones for their DPI and creates formal programs to support them. This might mean funding a maintainer, sponsoring a security audit, or dedicating staff time to contribute fixes.

If everyone free-rides on the commons, the commons eventually fails.

InnerSource and code reuse eliminate the waste of duplication. How many times has a citizen registration module, an authentication service, or a document management system been built independently by different ministries? The OSPO creates the platforms and culture for agencies to share and co-develop code, maintaining a catalogue of reusable components and saving millions in taxpayer funds.

Cross-border collaboration elevates the OSPO from a national function to a global actor (Figure 3 above).

- Finland’s municipalities benefit from a national procurement guide that gives even the smallest local authority practical tools for adopting open source. Finnish municipalities started adopting open source as early as 2011 (article).
- Kenya and Trinidad and Tobago are working with UN-backed OSPO initiatives to build new, sovereign digital infrastructure from the ground up, leapfrogging decades of technical debt.
- The EU’s Open Source Observatory and Repository, and the UN’s OSEE program, demonstrate that the challenges of digital sovereignty are shared, and that pooling resources to co-develop shared infrastructure costs each participant far less than building it alone.

Figure 5: Figure 4: Cost model comparison of proprietary vs open source

The True Cost of Open Source

Before governments can build an OSPO, they must discard one of the most persistent and damaging misconceptions in technology policy: that open source is free.

Open source is free as in *freedom*, not as in *cost*.

Confusing these two (freedom and cost) has led to under-resourced implementations, failed projects, and security disasters across governments worldwide. When budget decisions are made on the assumption of zero cost, programs are starved of the resources they need to succeed. The result is exactly the unmanaged risk that the OSPO was meant to address. The distinction becomes clear when you compare the two models (Figure 4).

In the proprietary model (Figure 4), money flows outward to external vendors for licenses, annual maintenance, support contracts, and upgrade fees. There are also hidden exit costs that vendors do not advertise. When you want to leave, you face data migration, process reengineering, and retraining costs that can equal or exceed the original software spend.

In the proprietary model, money leaves your government as you have built dependencies and not capabilities.

In the open source model (Figure 4), there are no license fees. Instead, you invest in seven categories:

1. **Internal staff** who operate and adapt systems
2. **Vendor support** for critical infrastructure (where needed)
3. **Training** that builds lasting workforce capability
4. **Security audits** that protect citizens and maintain compliance
5. **Custom development** that integrates systems and builds local industry and ecosystem
6. **Tooling and infrastructure** to see and manage the entire software stack
7. **Upstream contributions** that sustain the projects on which the government systems depend

The critical difference: **money stays in your economy.**

In the open source model, the assets you build, including skilled people, reusable code, and institutional knowledge, are yours. You own the result.

The main concern related to budget planning is not how much open source costs. Instead, the main concern should be focused on looking at where the money goes. That answer determines whether you are building dependency or building sovereignty.

Building a Government OSPO: A Phased Approach

It is not expected of any government (or an enterprise, for that matter) to build a mature OSPO overnight. The path forward is phased, evidence-driven, and deliberately modest at the start, a similar path that was followed by the most successful enterprise OSPOs.

- **Stage 1: Pathfinder:** A small team of two to three people that includes an OSPO lead, an engineer, and a policy specialist. They are typically embedded in an agency with a specific problem to solve. The goal is proof of concept.
- **Stage 2: Expansion:** In this stage, the OSPO team may grow to 5-8 staff with growing specialization supporting multiple agencies. Policy efforts move from pilot guidelines to government-wide requirements, with core services such as license scanning, SBOM management, and procurement training becoming repeatable processes. During this phase, the first strategic upstream contributions are made.
- **Stage 3: Maturity:** In this stage, the OSPO team may have ten to twenty or more specialists with a permanent budget line. The OSPO is now a strategic partner to every agency, an active contributor to global open source communities, and a catalyst for a thriving domestic technology ecosystem.

The governing principle across all three stages is simple: **budget follows maturity**.

Fund where you are on the maturity curve, not where you hope to be.

The Policy Foundation: Public Money, Public Code

At the heart of any government OSPO's policy framework should be a principle that is increasingly recognized around the world: **Public Money, Public Code**.

If the public pays for software to be built, that software is a public asset. It should be available for public use, to be reused by other agencies, scrutinized by citizens, and improved by the wider community.

This principle translates into three operational policies (Figure 5):

Figure 6: Figure 5: Operational policies

1. **Use-first:** before building or buying, look for an existing open source solution.
2. **Publish-by-default:** when you do build, publish the code, with exceptions justified explicitly (national security or similar).
3. **Contribute-first:** when your vendor modifies an open source component, you require that the improvement be contributed back to the upstream project, not kept as a private fork.

These three policies, embedded in law and enforced by a funded OSPO, transform a government's relationship with open source from passive, ad-hoc, and chaotic consumption to active stewardship. These policies are not new, as they have been adopted and have been proven in the enterprise world with high degrees of success.

The Choice Every Government Now Faces

The future of digital public infrastructure will be built on open source. This is a condition that already exists. The question of open source becoming foundational to public services was settled long ago at the infrastructure layer. The current open questions are how governments respond to that reality:

- Whether the infrastructure that supports citizen services is governed with intention or consumed with indifference.
- Whether that infrastructure becomes a source of sovereignty and resilience, or a source of exposure and dependency.
- Whether governments position themselves as responsible stewards of a global digital commons, contributing to it, sustaining it, and shaping its direction, or as passive riders, benefiting from the commons while investing nothing in its upkeep.

An OSPO is how a government answers those questions. The decision to establish a government OSPO is a deliberate choice to treat open source as a strategic asset. The OSPO models are proven from the enterprise's 25+ years of experience. The tools already exist, supported by a global community of practitioners who have already done much of the hard work of figuring out what works and are ready to share it.

No government starting this journey today needs to start from scratch. We don't need a perfect plan, just the institutional will to begin, and the discipline to build incrementally, measure progress, and invest adequately.

Governments that do this will own their digital future. Those who do not will continue renting it at increasing cost, with diminishing

control.

If you're interested in this topic, please reach out.

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